Three agents.
One pipeline.
No black boxes.

Multi-Agent Anomaly Detection
for Critical Infrastructure & Intelligence

ARGO	KEN	DEB
Data & Security	Detection	Briefing

SLINGSHOT 2026

Advanced Computing
& Intelligence

F1 = 1.00

Live integration test
22 May 2026

3 min 7 sec

Detection → Analyst briefing

4 domains

Zero architectural changes

THE PROBLEM

Critical systems generate signals. Most monitoring misses the threat until damage is done.

TOO SLOW

Conventional dashboards flag anomalies after the fact — after the thermal event, after the attack, after regulatory breach.

NO ATTRIBUTION

Alerts without explanation are operationally useless. In defence and regulated environments, an unexplained alert can be worse than no alert.

DOMAIN-LOCKED

Every monitoring system is built for one domain. Battery. Grid. OSINT. Re-architecture required every time. The threat doesn't respect domain boundaries.

0.273 mg/m³ — 5.5× the legal limit. A worker was breathing it. Underground. In the dark.

Marsham Edge detected it before the instruments agreed it was an anomaly.

THE SOLUTION

A coordinated team of three specialised AI agents.

RAW DATA — Any domain time-series → **ARGO** — Ingest · Validate · Provenance · Security → **KEN** — Detect · Classify · 4-trigger engine → **DEB** — Orchestrate · Brief · WhatsApp / dashboard → **ANALYST** — Structured report · Action-ready

ARGO

Data & Security

- Ingestion & schema validation
- Provenance tracking
- Adversarial robustness testing
- API surface monitoring
- Malformed record quarantine

KEN

ML & Detection

- 4-trigger hybrid detection engine
- Statistical + rate + geometric + ODE
- Physics-informed Trigger D
- Two-trigger gate (zero single-sensor noise)
- Domain-agnostic classifier

DEB

Orchestration & Briefing

- Task routing between Argo & Ken
- State machine management
- 5-point structured analyst briefing
- WhatsApp / dashboard delivery
- End-to-end latency: 3 min 7 sec

Four independent triggers.
Two must fire. Every alert is named.

A
Statistical Envelope

95% expanding-window envelope on rolling baseline.

Catches: Gradual drift outside established normal.

B
Rate Density Threshold

Sustained acceleration rate over rolling window.

Catches: Acceleration phase before failure.

C
Geometric Spike

Second-derivative inflection detection.

Catches: Abrupt sudden-onset events.

D
Physics-Informed ODE

First-principles residual from physics model.

Catches: Anomalies BEFORE B or C see acceleration.

KEY ADVANCE

TWO-TRIGGER GATE

Two independent triggers must fire simultaneously before any alert is issued.
Single noisy sensors cannot generate false alerts. Every alert names the specific triggers that fired.

PROVEN RESULTS

Live integration test.

22 May 2026. Real data. Real metrics.

AGENT	TASK	RESULT	KEY METRIC
ARGO	Malformed record quarantine (4 injected records)	4 / 4 detected 0 false quarantines	100% Precision
KEN	Campaign post detection 18 planted · 32 genuine	18 / 18 detected 0 false positives	F1 = 1.00
DEB	5-point analyst briefing via WhatsApp	5 / 5 delivered full pipeline	3 min 7 sec

F1 = 1.00

Ken · OSINT detection■Precision & Recall both 1.00

3:07

Deb · End-to-end■detection to analyst briefing

0

False positives■across all 32 genuine records

Test dataset: 54 OSINT records · 18 planted campaign posts · 4 malformed records · 32 genuine records

All three agents ran sequentially without human intervention. Deb delivered the briefing while the pipeline was still completing.

DOMAIN VERSATILITY

Same three agents.
Any domain. Zero architectural change.

1	2	3	4
Li-ion Battery Safety	**Electricity Markets**	**Occupational Health**	**OSINT / Adversarial**
Thermal anomaly detection Live MVP at liguard.streamlit.app Real-time four-trigger state machine	Singapore NEMS forecasting 0.9% MAPE · 35,424 real records Independently validated 2025	Silica dust sensor fusion 174 readings · 31 exceedances 0.273 mg/m³ — 5.5× legal limit	Influence campaign detection F1 = 1.00 · 18/18 posts found 0 false positives on live data

The platform is domain-agnostic by design. If it generates time-series data, Marsham Edge can monitor it.

Deployment: Docker · NanoClaw v2.0.64 · Air-gap capable · No cloud AI dependency

ACADEMIC VALIDATION

Not claimed. Published.

35,424 real records. Independently validated. 2025.

Independent Academic Research · Singapore Electricity Market · Project #2420-0002

Singapore NEMS real-market dataset · March 2025 · 35,424 records

0.9%	3.28%	95.24%
Prophet Demand Forecast MAPE (1-day horizon)	CNN-LSTM Price Forecast MAPE vs 6.28% standalone LSTM	Random Forest Spike Classification CV Accuracy

WHY THIS MATTERS FOR SLINGSHOT JUDGES

Independent academic validation on real-world data replaces vendor claims. This research was not commissioned as a marketing exercise.

The same architecture validated on 35,424 Singapore electricity market records now runs on battery safety, OSINT, and occupational health —

with no architectural change. Generalisation to new domains is demonstrated, not projected.

MARKET OPPORTUNITY

The market is large.

The timing is right. The regulation is forcing it.

USD 9.7B	USD 15B+	USD 25B+
Anomaly Detection Market by 2028 (CAGR 20%)	Defence AI & Critical Infrastructure Intelligence	Combined Addressable Market

Sovereign AI mandates

Singapore, Australia, Japan, Gulf — all requiring on-premise or air-gap deployable AI. Marsham Edge is sovereign-ready on day one.

Regulatory pressure

OHS, energy market, defence procurement all requiring explainable, auditable AI decisions. Black boxes are becoming a legal liability.

Multi-domain operations

Defence and critical infrastructure operators are converging — one operator needs battery + grid + OSINT + worker safety. That is exactly Marsham Edge.

Three revenue streams.
One proof point that removes the sales cycle.

SaaS Licence

Per-domain deployment
Annual subscription
From USD 2,500/month

Professional Services

Domain onboarding
Sensor integration
Trigger calibration
USD 15K–40K / deployment

Sovereign / Enterprise

Perpetual licence
Air-gap deployments
Defence / government
USD 150K–500K

Live demo
liguard.streamlit.app — upload any CSV, run the full four-trigger engine. No login. No sales call. Judges qualify themselves.

Partnerships
High Speed Rail Authority Australia · Schneider Electric · RATP — enterprise relationships, not name-drops.

Competitions
Cap Vista / MINDEF-AMIAD · UCWS Singapore 2026 (Agent + Deep Research tracks) · Slingshot 2026 — defence and enterprise procurement pipelines.

Research Interns
Active academic collaboration embedding ML researchers directly in Ken and Argo development.

THE TEAM

Muriel Demarcus

Founder & CEO, Marsham Edge Pte. Ltd.

MSc, Ecole Centrale Paris (Paris-Saclay University)
30+ years: multi-billion-dollar infrastructure and digital transformation, Asia-Pacific
Domains: defence, energy, critical infrastructure, transport, occupational health
Based: Singapore

Research Interns

Active ML researchers embedded in Ken and Argo development — time-series anomaly detection and computer vision. Every engagement, every project.

NOT THE USUAL FOUNDER

Female. 52. French. Ultrarunner. Engineer. 30 years before building this. The platform was built because the problem is real - not because it was fundable.

WHAT WE'RE ASKING

Bring us your data.
Meet Deb.

Live demo now

liguard.streamlit.app
Upload any CSV with timestamp + sensor column.
The full four-trigger state machine runs in real time.

If selected

We will bring a real dataset to Demo Day.
Deb will coordinate Argo and Ken live.
You time the briefing.

Infrastructure

Self-hosted · Air-gap deployable
Docker · NanoClaw v2.0.64
No cloud AI dependency

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